

**The Reachable Human Phenotype:
Factorized Selfhood, Morphology-Congruent Binding, and
What the Decisive Experiment Costs**

PIATRA . INSTITUTE

August 2026

Abstract

Capability, body ownership, agency and identity are studied in separate literatures, and each has shown that its object can be moved: synchronous multisensory evidence produces ownership of rubber hands, third arms and virtual cat ears, avatars alter later behavior, and some communities report persistent nonhuman self-models or cultivated inner agents. We treat these findings as coordinates of one object, the reachable human phenotype, the set of configurations a nervous system can enter through physiological change, sensory perturbation, learning and context, and propose ten coordinates. A single scalar would miss the clearest available dissociation, in which ownership of virtual cat ears rose in 24 participants while proprioceptive drift did not. The one direct study of body ownership in a nonhuman-identifying community found 50 furries reporting weaker embodiment of a human rubber hand, the opposite of what global plasticity predicts. A causal-inference model of ownership with a morphology-congruence prior reproduces that sign and, from the same parameter shift, predicts ownership of a congruent nonhuman body higher by 0.22. In simulation, the design used so far recovers the true theory in 0.51 of fits and reads morphology-congruent binding as global plasticity 96.5 percent of the time; a crossed design manipulating morphology against identity congruence recovers it in 0.94 and needs about 25 participants per group for 80 percent power on the diagnostic interaction. A multiscale formalism separates fast state from slowly learned parameters, makes hysteresis measurable (55 percent of a trained gain survives the end of the perturbation), and states the SELF and OTHER binding hypothesis as a crossed prediction. None of the results is evidence about any person or community.

Introduction

An ordinary waking adult has one body, one set of capabilities and one apparent author of their thoughts. Each of these appears to be a fixed property of the organism, and each has been altered in the laboratory. Ownership of a rubber hand follows synchronous visuotactile stimulation (Botvinick and Cohen 1998). Ownership extends to a third arm (Guterstam et al. 2011), to supernumerary robotic limbs whose incorporation appears in peripersonal-space measures as well as questionnaires (Arai et al. 2022), and to avatars with nonhuman morphology that participants learn to control (Won et al. 2015; Krekhov et al. 2019). Avatars also alter later behavior: people given taller or more attractive virtual bodies behave differently afterward (Yee and Bailenson 2007).

If the phenotype moves along this many axes, the ordinary configuration is one point in a larger space. We call that space the reachable human phenotype, the set of behavioral and phenomenological configurations a nervous system can enter through physiological change, sensory perturbation, learning and context. Two questions follow: what its coordinates are, and what it would cost to determine which of them move independently. We answer the second with a sample size.

Coordinates of the phenotype

We propose ten coordinates: capability, body ownership, body schema, self-location, agency, identity, memory access, perceptual organization, affect and reality testing. Factorization has precedent. Bodily self-consciousness already decomposes into ownership, self-location and first-person perspective, which dissociate under different manipulations (Blanke et al. 2015), and the owning self is itself inferred, partly from interoceptive prediction (Seth 2013).

The strongest reason to keep the coordinates separate is a 2026 experiment in which 24 participants received virtual cat ears with congruent visual, tactile and auditory manipulation. Subjective ownership of the ears rose and touch and sound were referred to them, while proprioceptive drift did not change (Yamamura et al. 2026). Ownership changed without evidence that the ears had become a spatially grounded part of the body schema. A study the same year found that active and passive haptics increased the sense of embodiment of virtual dog ears in 28 participants (Khan et al. 2026). A single scalar measure of alternate embodiment would have hidden the cat-ear dissociation, the clearest structural result available. A meta-analysis of 111 articles also reports modest ownership effects, somewhat stronger agency effects, and widespread problems with power and inconsistent terminology (Mottelson et al. 2023).

Evidence from nonhuman-identifying and plural communities

Three communities are relevant. Therians identify as or with a nonhuman animal; furry fandom involves anthropomorphic animal characters, and a fursona often functions as a self-representation; tulpamancers deliberately cultivate an inner agent experienced as having its own perspective. These are communities with their own vocabularies and are not clinical categories, and nothing below infers disorder from unusual content.

The quantitative record is thin. In 112 therians and 265 comparison participants, unusual experiences, introverted anhedonia and autism-quotient scores predicted therian status, with odds of exceeding the autism-quotient cutoff about 5.9 times higher and a wide interval from 2.04 to 17.31 (Clegg et al. 2019). The design is cross-sectional and the causal direction unresolved. Phenomenological work documents phantom body parts and shifts in 5 in-depth interviews (Grivell et al. 2014), and a 2025 phenomenological study describes these experiences as involving body, perception, environment, affect and behavior in addition to belief (Luiggi-Hernández et al. 2025). Liking an animal and identifying as one are psychometrically separable, as a self-binding account requires (Roberts et al. 2015). Among 243 participants across tulpamancy, ASMR and comparison groups, tulpamancers showed unusual sensory experience without a general elevation in delusion-proneness (Palmer-Cooper et al. 2021), and experienced autonomy of characters also occurs among adult fiction writers (Taylor et al. 2003). Clinical therianthropy is a separate phenomenon with its own review, whose authors caution against equating nonclinical identities with clinical presentations; we cite it only as a boundary condition (Blom and Sharpless 2025).

No study has compared these groups under matched embodiment and agency manipulations, and no therian-specific or tulpamancer-specific neuroimaging, stimulation or longitudinal training ev-

idence establishes a mechanism. Most of what follows concerns the design of an experiment that would fill this gap.

A morphology-congruence model of body ownership

A simple account holds that members of nonhuman-identifying communities have more plastic body models in general and should therefore show stronger illusions. The one direct study found the opposite. Fifty furies reported weaker subjective embodiment of a human rubber hand than comparison data, and within the sample, participants who valued the human form less showed less proprioceptive drift toward the human hand (Kranjec et al. 2019). The study uses previously collected comparison data without a contemporaneous control group and appears in conference proceedings, and both limitations apply wherever it is used. Taken at face value, it contradicts the global-plasticity account.

A morphology-congruent account predicts this result. Body ownership can be modeled as inference about whether visual and proprioceptive evidence share a common bodily cause (Samad et al. 2015). We add a prior over morphological correspondence, so that the readiness with which an effector is taken as part of the body depends on how well its form matches the participant's self-model. Shifting this one prior for an identifying group, with less prior mass on a human effector and more on a congruent nonhuman one, lowers ownership of the human hand by 0.22 and raises ownership of the congruent nonhuman body by 0.22, leaving a neutral nonhuman morphology unchanged (Figure 1, left). The same parameter produces both signs, and the prediction is testable: participants who bind a human hand less should bind a congruent nonhuman body more.

The model also reproduces the cat-ear dissociation. When ownership is read as the posterior probability of a common cause and drift as the resulting shift in felt position, a change in the prior moves both, in proportions that depend on the visual-proprioceptive discrepancy. At small discrepancy the ownership change dominates and drift barely moves; at large discrepancy drift dominates (Figure 1, right). Ownership without drift is therefore expected in an identifiable region of parameter space, and the null drift result locates the experiment within it.

Design analysis

Three theories are under consideration. Global plasticity holds that identifying participants embody every body more readily. Morphology-congruent binding holds that they bind bodies matching their self-model more readily and human bodies less. Expectancy holds that the differences are confined to subjective report and track awareness of the hypothesis, a serious possibility given that rubber-hand reports respond to demand characteristics and that ownership and drift dissociate. The question is which experiment can distinguish them.

We simulated data from each theory in turn and fitted all three by model selection, under the design used so far and under richer alternatives (Figure 2). With a single human effector, recovery is 0.51 against a chance level of 0.33. When morphology-congruent binding is true, that design selects global plasticity 96.5 percent of the time. The one direct study uses this design, so it could not

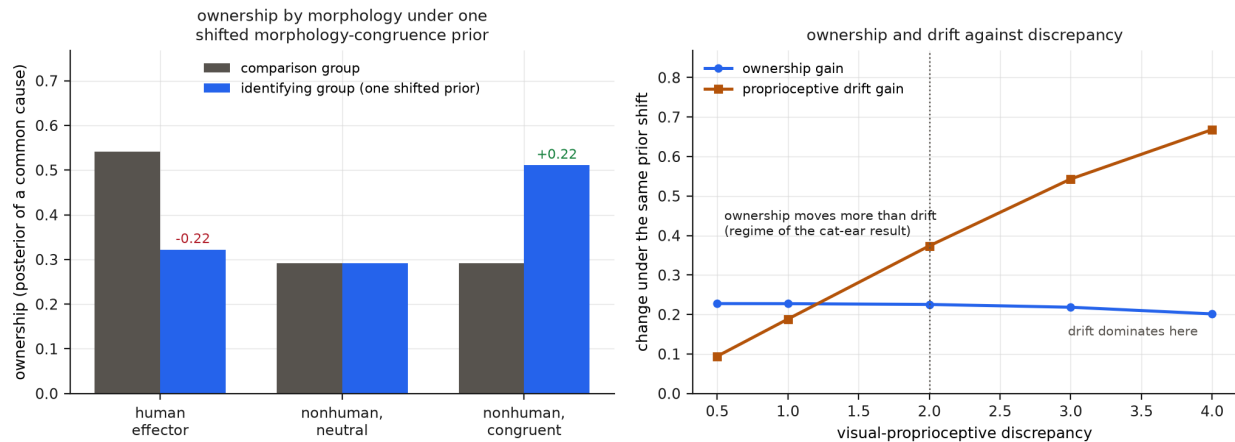


Figure 1: Morphology-congruent binding. Left: a single shifted prior over morphological correspondence lowers ownership of a human effector and raises it for a congruent nonhuman one by the same amount, reproducing the observed sign and predicting its mirror. Right: under the same shift, ownership and proprioceptive drift change in proportions that depend on the sensory discrepancy, so ownership without drift occupies a definite regime.

have distinguished the account it appears to refute from the account it supports. Adding a generic nonhuman avatar raises recovery to 0.73 but identifies congruent binding only 41 percent of the time, because a neutral nonhuman body is the condition in which the two theories agree. The crossed design, varying morphology against identity congruence within participants, reaches 0.94 and recovers congruent binding 97.5 percent of the time.

The diagnostic quantity is an interaction, so the sample must be sized for the interaction. At the modeled effect, 20 participants per group give power of 0.71 and about 25 per group give 80 percent, and more are needed once several groups and several outcome families enter. Embodiment experiments typically use about 20. The protocol's decisive test is an identity-congruent interaction in at least one subjective measure and at least one behavioral, physiological or psychophysical measure, since a questionnaire difference alone cannot distinguish a changed body model from a changed expectation.

A multiscale formalism

A rubber-hand illusion develops in seconds and a fursona over years, so the model needs more than one timescale. The formalism separates a fast neural and physiological state driven by perturbation, stable individual traits, and slowly learned generative-model parameters that accumulate from the fast state and decay when it stops. The safely reachable phenotype is the set of configurations attainable under bounded, ethically permissible perturbations within a horizon, a reachable set in the control-theoretic sense with safety imposed as a hard constraint.

Three quantities become measurable. Hysteresis, the fraction of a trained configuration that survives after the driving perturbation stops, is 55 percent of the trained gain in the model; it distinguishes a learned identity, which persists, from a maintained illusion, which collapses with its



Figure 2: Design discrimination. The three matrices show which theory model selection picks when each is true, under the design run so far, under a generic-nonhuman variant, and under the crossed design; the human-only design reads morphology-congruent binding as global plasticity 96.5 percent of the time. Right: power for the diagnostic interaction against sample size, with the coarse grid and a per-participant scan near the 80 percent point.

stimulus. Learning rates for binding a representation to SELF and to OTHER can be estimated separately, which makes the binding hypothesis quantitative. And the region reachable under training exceeds the fast-driven baseline by a factor of 2.7 in the model, so most of the phenotype space of interest cannot be reached in a single session and requires longitudinal designs.

The generative-binding hypothesis can now be stated as a prediction. Therian-like identification may involve morphology-congruent binding of nonhuman representations to SELF; cultivated inner agents may involve learned binding of internally generated representations to OTHER. Under matched training with a novel creature to which no participant has prior attachment, the hypothesis predicts a crossed sign: faster SELF binding in one group, faster OTHER autonomization in the other, and the comparison group at neither extreme (Figure 3, left). General suggestibility would produce a main effect, which the hypothesis excludes. Under the modeled learning-rate gap, the crossing reaches 80 percent power at about 11 participants per group by the closed-form two-sample calculation, and simulated power at 15 per group is 0.90, so a modest study could refute it.

Limits of the software metaphor

Such inquiries are often motivated by the idea that the self is software and that unusual identities are a configuration loaded incorrectly. The metaphor is correct that the adult self is not read directly off the biology and that some process determines what counts as body, self and agent. It is wrong in four respects. It implies a single initialization, whereas the evidence indicates continuous recalibration from sensorimotor history, interoception, social feedback and current state. It implies a discrete species template, which cannot account for identities whose referents are mythological or invented, and which are better located at semantic and autobiographical levels than in a library of morphologies. It conflates bodily ownership with narrative identity, although a person can enjoy an avatar, feel represented by it, report phantom sensations and describe themselves in its

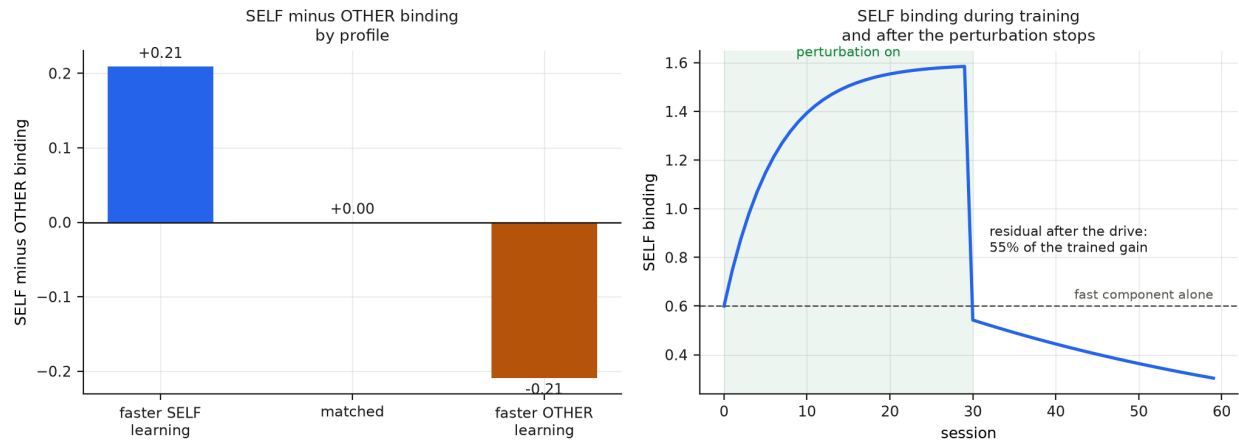


Figure 3: The multiscale reachable set. Left: the SELF and OTHER hypothesis predicts a crossed sign across profiles, which distinguishes it from general suggestibility. Right: hysteresis, with 55 percent of the trained gain surviving after the perturbation stops, which distinguishes a learned configuration from a maintained illusion.

terms without these being one psychological operation. And it equates difference with malfunction, although a persistent nonstandard self-model may be neutral, valuable or distressing, and function must be measured. The terms used here, morphology-congruent binding, agent-model autonomization, hysteresis and reachable-set mapping, are less evocative and more testable.

Natural experiments: fever in autism

Populations that already show unusual transitions provide starting coordinates, and the fever literature shows how to report such a case with its limits. An early prospective study of 30 febrile and 30 afebrile autistic children found transient reductions on several aberrant-behavior measures during fever (Curran et al. 2007). A later prospective study of 141 autistic children found that of the 50 who had a fever, 3 showed reported improvement, and the typical pattern was worsening (Byrne et al. 2022). Fever-associated improvement may therefore offer a rare reversible natural experiment in state-dependent behavioral access; it is uncommon, and it does not support the claim that fever unlocks dormant capacities. The value of such a case lies in its reversible structure, from baseline to altered state to recovery, because reversibility distinguishes a state transition from a stable trait difference.

Ethics and dual use

The communities discussed here are not a resource for cognitive science, and three commitments follow. Terminology belongs to communities, and we use their terms as they use them; cognitive interviews should precede measurement, so that a spiritual identity, a role, a self-representation, a bodily sensation and a literal biological belief are not collapsed into one variable called animal identification. Difference is not disorder: the clinical literature enters only as a boundary condition, with its authors' warning attached, distress and impairment are measured, and no finding about a

group licenses an inference about an individual. Participants recruited from small online communities are re-identifiable in ways that general-population samples are not, which constrains data sharing and case description. Four further commitments apply: preregistered primary outcomes, measurement of expectancy, community consultation without giving any party control over conclusions, and publication of null results. Closed-loop exploration, in which an algorithm proposes the next perturbation to maximize expected information about the phenotype, requires the safety constraint to be hard, and any novel state it reports requires within-person and independent replication before it is treated as a discovery and not a measurement artifact.

Objections

The strongest objection is that the simulations contain no people, so the numbers are properties of assumptions. This is correct and is stated wherever a number appears. The recovery matrices and sample sizes are claims about what a given experiment could resolve under a stated effect, and such claims belong before participants are recruited. A second objection is that a framework spanning fever, avatars and identity is too broad to constrain anything. At the level of mechanism it is: fever involves immune and metabolic state, embodiment involves multisensory inference, and cultivated agents involve learned source attribution. The phenomena share a method, repeated within-person measurement under reversible perturbation with factorized outcomes, and probably no mechanism. A third objection is that studying communities as unusual cases risks pathologizing them. The risk is real. In the framework these populations demonstrate what the ordinary nervous system can do, and the design analysis guards most strongly against reading a questionnaire difference as a difference in bodily computation. A fourth objection is that the one direct study is a proceedings paper with historical controls. We use it only for a sign that the model reproduces, and recommend that it be repeated with contemporaneous controls in the crossed design.

Falsification

The framework fails if the coordinates collapse: if ownership, body schema, agency, identity and reality testing load on one factor across manipulations, the factorization adds nothing and the cat-ear dissociation would have to be a measurement failure. The morphology-congruent account fails if group differences appear equally across every morphology, the global-plasticity signature the crossed design is built to detect. The SELF and OTHER hypothesis fails if both communities show the same undifferentiated effect, since a shared main effect is what suggestibility predicts. The learned-autonomization account fails if longitudinal training produces no change beyond practice effects, and the hysteresis claim fails if trained configurations vanish when the manipulation stops. The trait account fails if group differences vanish after adjustment for expectancy and imagery, which would place the phenomenon in report and not in body computation. The design analysis itself fails if a well-powered crossed experiment recovers the theories no better than the human-only design did in simulation, which would mean the generative models misdescribe where the information lies.

Scope of the claims

No mechanism is established, no community is characterized and no capability is unlocked. The proposal is that capability, embodiment, agency and identity are coordinates of one measurable object; that the object is state-dependent and only partly explored; that the phenomena that look strangest from outside occupy some of its most informative regions; and that the experiment that would settle the current question is crossed and needs about 25 participants per group, compared with the 20 typical of the field, before multiple groups and outcome families are added.

Reproducibility

The simulation, its 3 figures and every model number quoted above regenerate from one command in the repository (`simulation/run_all.py`; seeded, bit for bit). 22 invariants fail the run if broken, among them the human-only design's failure to discriminate and its misreading of congruent binding, the crossed design's recovery, the monotone power curve, the per-participant scan bracketed by the coarse grid, the closed-form sample size for the SELF and OTHER crossing, the two-sign property with its neutral-morphology control, the ownership-over-drift regime at small discrepancy and its reversal at large, positive but incomplete hysteresis, the crossed form of the double dissociation, and the trained region exceeding the fast-driven baseline. Every quantity is a property of the stated generative models and none is an empirical estimate; externally measured figures quoted in the text enter `results.json` as cited records with their sources.

References

- Arai, Ken, Hiroto Saito, Masaaki Fukuoka, et al. 2022. "Embodiment of Supernumerary Robotic Limbs in Virtual Reality." *Scientific Reports* 12: 9769. <https://doi.org/10.1038/s41598-022-13981-w>.
- Blanke, Olaf, Mel Slater, and Andrea Serino. 2015. "Behavioral, Neural, and Computational Principles of Bodily Self-Consciousness." *Neuron* 88 (1): 145–66. <https://doi.org/10.1016/j.neuron.2015.09.029>.
- Blom, Jan Dirk, and Brian A. Sharpless. 2025. "A Systematic Review on Clinical Therianthropy and a Proposal to Conceptualize Zoomorphism as a Diagnostic Spectrum." *Neuroscience and Biobehavioral Reviews* 174: 106193. <https://doi.org/10.1016/j.neubiorev.2025.106193>.
- Botvinick, Matthew, and Jonathan Cohen. 1998. "Rubber Hands 'Feel' Touch That Eyes See." *Nature* 391 (6669): 756. <https://doi.org/10.1038/35784>.
- Byrne, Katherine, Shuting Zheng, Somer Bishop, et al. 2022. "Behavioral Responses to Fevers and Other Medical Events in Children With and Without ASD." In *medRxiv*, Preprint 2022.05.23.22275374. <https://doi.org/10.1101/2022.05.23.22275374>.
- Clegg, Helen, Roz Collings, and Elizabeth C. Roxburgh. 2019. "Therianthropy: Wellbeing, Schizotypy, and Autism in Individuals Who Self-Identify as Non-Human." *Society and Animals* 27 (4): 403–26. <https://doi.org/10.1163/15685306-12341540>.

- Curran, Laura K., Craig J. Newschaffer, Li-Ching Lee, Stephen O. Crawford, Michael V. Johnston, and Andrew W. Zimmerman. 2007. "Behaviors Associated With Fever in Children With Autism Spectrum Disorders." *Pediatrics* 120 (6): e1386–92. <https://doi.org/10.1542/peds.2007-0360>.
- Grivell, Timothy, Helen Clegg, and Elizabeth C. Roxburgh. 2014. "An Interpretative Phenomenological Analysis of Identity in the Therian Community." *Identity* 14 (2): 113–35. <https://doi.org/10.1080/15283488.2014.891999>.
- Guterstam, Arvid, Valeria I. Petkova, and H. Henrik Ehrsson. 2011. "The Illusion of Owning a Third Arm." *PLOS ONE* 6 (2): e17208. <https://doi.org/10.1371/journal.pone.0017208>.
- Khan, Omar A., Bao Han Trinh, Lee Lisle, and Tiffany D. Do. 2026. "Having Dog Ears 'for Real': Effects of Active and Passive Haptics on Embodying Non-Human Body Parts in VR." In *arXiv*, arXiv:2606.26364. Preprint. <https://doi.org/10.48550/arXiv.2606.26364>.
- Kranjec, Alexander, Louis Lamanna, Erick Christopher Guzman, et al. 2019. "Illusory Body Perception and Experience in Furies." *Proceedings of the 41st Annual Meeting of the Cognitive Science Society*, 596–602. <https://escholarship.org/uc/item/37x5162v>.
- Krekhov, Andrey, Sebastian Cmentowski, and Jens Krüger. 2019. "The Illusion of Animal Body Ownership and Its Potential for Virtual Reality Games." *2019 IEEE Conference on Games (CoG)*, 1–8. <https://doi.org/10.1109/CIG.2019.8848005>.
- Luiggi-Hernández, José G., Elizabeth Fein, Jennifer Bradley, Jenna Pelly, Sharon E. Roberts, and Kathleen C. Gerbasi. 2025. "The Jackal in the City: An Empirical Phenomenological Study of Embodied Experience Among Therians and Otherkin." *The Humanistic Psychologist* 53 (1): 40–58. <https://doi.org/10.1037/hum0000350>.
- Mottelson, Aske, Andreea Muresan, Kasper Hornbæk, and Guido Makransky. 2023. "A Systematic Review and Meta-Analysis of the Effectiveness of Body Ownership Illusions in Virtual Reality." *ACM Transactions on Computer-Human Interaction* 30 (5): 1–42. <https://doi.org/10.1145/3590767>.
- Palmer-Cooper, Emma, Nicola McGuire, and Abigail Wright. 2021. "Unusual Experiences and Their Association with Metacognition: Investigating ASMR and Tulpamancy." *Cognitive Neuropsychiatry* 27 (2-3): 86–104. <https://doi.org/10.1080/13546805.2021.1999798>.
- Roberts, Sharon E., Courtney N. Plante, Kathleen C. Gerbasi, and Stephen Reysen. 2015. "The Anthrozoomorphic Identity: Furry Fandom Members' Connections to Nonhuman Animals." *Anthrozoös* 28 (4): 533–48. <https://doi.org/10.1080/08927936.2015.1069993>.
- Samad, Majed, Albert Jin Chung, and Ladan Shams. 2015. "Perception of Body Ownership Is Driven by Bayesian Sensory Inference." *PLOS ONE* 10 (2): e0117178. <https://doi.org/10.1371/journal.pone.0117178>.
- Seth, Anil K. 2013. "Interoceptive Inference, Emotion, and the Embodied Self." *Trends in Cognitive Sciences* 17 (11): 565–73. <https://doi.org/10.1016/j.tics.2013.09.007>.

- Taylor, Marjorie, Sara D. Hodges, and Adèle Kohányi. 2003. "The Illusion of Independent Agency: Do Adult Fiction Writers Experience Their Characters as Having Minds of Their Own?" *Imagination, Cognition and Personality* 22 (4): 361–80. <https://doi.org/10.2190/FTG3-Q9T0-7U26-5Q5X>.
- Won, Andrea Stevenson, Jeremy Bailenson, Jimmy Lee, and Jaron Lanier. 2015. "Homuncular Flexibility in Virtual Reality." *Journal of Computer-Mediated Communication* 20 (3): 241–59. <https://doi.org/10.1111/jcc4.12107>.
- Yamamura, Hiroo, Ryota Kondo, Kuniharu Sakurada, Jean-Marie Normand, and Maki Sugimoto. 2026. "The Imaginary Cat Ears Illusion: An Analysis of Ownership for an Avatar's Cat Ears via Visual–Auditory–Haptic Multimodal Feedback in Virtual Reality." *Frontiers in Virtual Reality* 7: 1817800. <https://doi.org/10.3389/frvir.2026.1817800>.
- Yee, Nick, and Jeremy Bailenson. 2007. "The Proteus Effect: The Effect of Transformed Self-Representation on Behavior." *Human Communication Research* 33 (3): 271–90. <https://doi.org/10.1111/j.1468-2958.2007.00299.x>.